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Faculty of Mechanical and Industrial Engineering

Fluid Mechanics (MEng3133)

Chapter Five

Dimensional Analysis and Similarity

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Outline

1. Introduction
2. Dimensional Homogeneity
3. Buckingham Pi Theorem
4. Non Dimensionalization of Basic Equations
5. Similitude
6. Significance of Non-Dimensional Numbers In Fluid Flows



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- Dimensional analysis is a method for **reducing the number** and **complexity** of experimental variables that affect a given physical phenomenon, by using a sort of compacting technique.
- Significance of dimensional variables;
 - ❖ The first is enormous savings in time and money.
 - ❖ it helps our thinking and planning for an experiment or theory.
 - ❖ provides scaling laws that can convert data from a cheap, small model to design information for an expensive, large prototype.
- In fluid mechanics, the four basic dimensions are usually taken to be mass M , length L , time T , and temperature Θ , or an **MLT Θ** system for short.



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Primary dimensions

mass	M	(kilogram, kg)
length	L	(metre, m)
time	T	(second, s)
temperature	Θ	(kelvin, K)
electric current	I	(ampere, A)
luminous intensity	C	(candela, cd)
amount of substance	n	(mole, mol)

In the following, [] means “dimensions of”.

Use the definition to determine the dimensions of viscosity

$$\tau = \mu \frac{dU}{dy}$$

From the definition,

$$\mu = \frac{\tau}{dU/dy} = \frac{\text{force / area}}{\text{velocity / length}}$$

Hence,

$$[\mu] = \frac{MLT^{-2}/L^2}{LT^{-1}/L} = ML^{-1}T^{-1}$$



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	Quantity	Common Symbol(s)	Dimensions
Geometry	Area	A	L^2
	Volume	V	L^3
	Second moment of area	I	L^4
Kinematics	Velocity	U	LT^{-1}
	Acceleration	a	LT^{-2}
	Angle	θ	1 (i.e. dimensionless)
	Angular velocity	ω	T^{-1}
	Quantity of flow	Q	L^3T^{-1}
	Mass flow rate	$\dot{m}$	MT^{-1}
Dynamics	Force	F	MLT^{-2}
	Moment, torque	T	ML^2T^{-2}
	Energy, work, heat	E, W	ML^2T^{-2}
	Power	P	ML^2T^{-3}
	Pressure, stress	p, τ	$ML^{-1}T^{-2}$
Fluid properties	Density	ρ	ML^{-3}
	Viscosity	μ	$ML^{-1}T^{-1}$
	Kinematic viscosity	ν	L^2T^{-1}
	Surface tension	σ	MT^{-2}
	Thermal conductivity	k	$MLT^{-3}\Theta^{-1}$
	Specific heat	c_p, c_v	$L^2T^{-2}\Theta^{-1}$
	Bulk modulus	K	$ML^{-1}T^{-2}$



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- Since $Re = UL\rho/\mu$ is known to be dimensionless, the dimensions of μ must be the same as those of ρUL ;

$$[\mu] = \frac{MLT^{-2}/L^2}{LT^{-1}/L} = ML^{-1}T^{-1}$$



The Principle of Dimensional Homogeneity (PDH)

“If an equation truly expresses a proper relationship between variables in a physical process, it will be **dimensionally homogeneous**; that is, each of its additive terms will have the same dimensions.”

- ❖ **Dimensional variables** are the quantities that actually vary during a given case and would be plotted against each other to show the data.
- ❖ **Dimensional constants** may vary from case to case but are held constant during a given run. They are normally used to help non-dimensionalize the variables in the problem. V_0, s_0, g, ρ, c
- ❖ **Pure constants** have no dimensions and never did. Angles and revolutions are dimensionless. e, π



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- Note that integration and differentiation of an equation may change the dimensions but not the homogeneity of the equation.
- The **variables** are the things we wish to plot, the basic output of the experiment or theory: in this case, S versus t . The **parameters** are those quantities whose effect on the variables we wish to know: in this case S_0 , V_0 , and g .

$$S = S_0 + V_0 t + \frac{1}{2} g t^2 \qquad \frac{p}{\rho} + \frac{1}{2} V^2 + g z = \text{const}$$

- Among our parameters, we therefore select **scaling parameters (also called repeating variables)**, used to define dimensionless variables.



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- **Selection of Scaling (Repeating) Variables**
 1. They must not form a dimensionless group among themselves, but adding one more variable will form a dimensionless quantity.
 2. Do not select output variables for your scaling parameters.
 3. If convenient, select popular, not obscure, scaling variables because they will appear in all of your dimensionless groups.



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For the free falling we have three options.

- **Option 1**; Scaling parameters S_0 and V_0 : the effect of gravity g .

$$S^* = \frac{S}{S_0} \quad t^* = \frac{V_0 t}{S_0}$$

$$S^* = 1 + t^* + \frac{1}{2}\alpha t^{*2} \quad \alpha = \frac{gS_0}{V_0^2} \quad (5.10)$$

- **Option 2**: Scaling parameters V_0 and g : the effect of initial displacement S_0 ;

$$S^{**} = \frac{Sg}{V_0^2} \quad t^{**} = t \frac{g}{V_0}$$

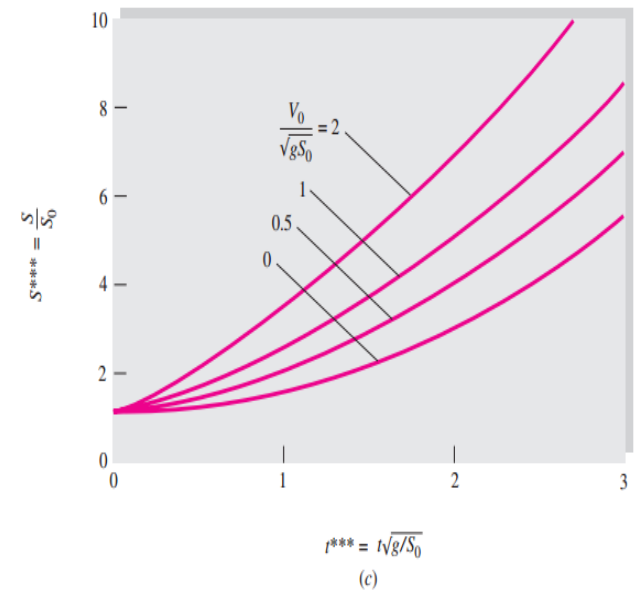
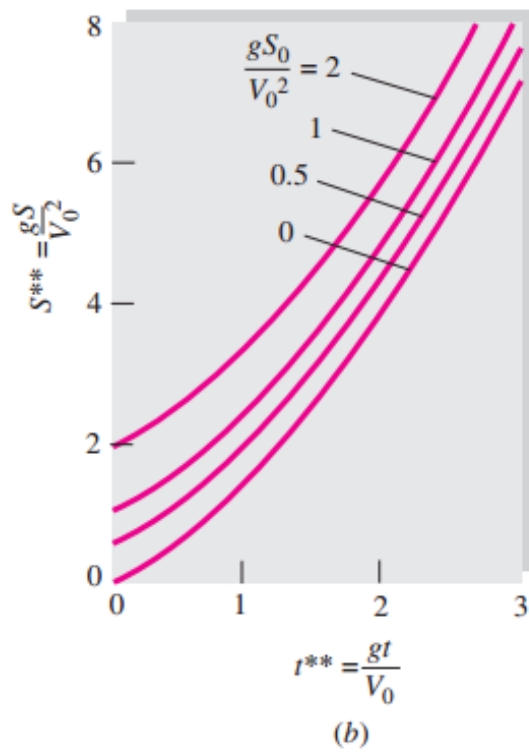
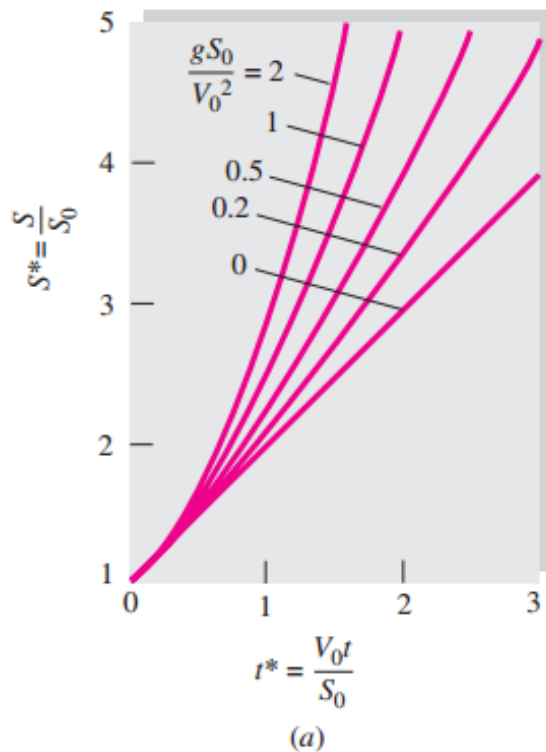
$$S^{**} = \alpha + t^{**} + \frac{1}{2}t^{**2} \quad \alpha = \frac{gS_0}{V_0^2}$$



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Option3: Scaling parameters S_0 and g : the effect of initial speed V_0 ;

$$S^{***} = \frac{S}{S_0} \quad t^{***} = t \left(\frac{g}{S_0} \right)^{1/2} \quad S^{***} = 1 + \beta t^{***} + \frac{1}{2} t^{***2} \quad \beta = \frac{1}{\sqrt{\alpha}} = \frac{V_0}{\sqrt{g S_0}}$$





The Buckingham Pi Theorem

“If a physical process satisfies the PDH and involves n dimensional variables, it can be reduced to a relation between only k dimensionless variables or Pi's. The reduction $j = n - k$ equals the maximum number of variables that do not form a pi among themselves and is always less than or **equal to the number of dimensions** describing the variables”

$j = n - k$ where n = dimensional variables

$j \leq$ number of dimensions (scaling variables)

k = dimensionless variables (II's)

Note :- On rare occasions it may take more pi groups than this minimum



Exponent Method (PI theorem)

- **Six steps to find Pi group;**
 1. List and count the n variables involved in the problem. If any important variables are missing, dimensional analysis will fail.
 2. List the dimensions of each variable according to $MLT\Theta$.
 3. Find j . Initially guess j equal to the number of different dimensions present, and look for j variables that do not form a pi product. If no luck, reduce j by 1 and look again.
 4. Select j scaling parameters that do not form a pi product.
 5. Add one additional variable to your j repeating variables, and form a power product. Algebraically find the exponents.
 6. Write the final dimensionless function, and check.



- Example1;** Suppose one knew that the force F on a particular body shape immersed in a stream of fluid depended only on the body length L , stream velocity V , fluid density ρ , and fluid viscosity μ ; that is,

$$F = f(L, V, \rho, \mu)$$

$$\frac{F}{\rho V^2 L^2} = g\left(\frac{\rho V L}{\mu}\right)$$

$$C_F = g(\text{Re})$$

develop the dimensional groups, using PI theorem;

Solution;

Step1; Write the function and count variables: $F = f(L, V, \rho, \mu)$ 5 variables.

Step2; List dimensions of each variable

F	L	U	ρ	μ
$\{MLT^{-2}\}$	$\{L\}$	$\{LT^{-1}\}$	$\{ML^{-3}\}$	$\{ML^{-1}T^{-1}\}$

Step3; find j ; $j=3$, $n-k=j$, $k=2$ (pi groups), j is number of dimensions

Step4; Select repeating j variables. The group L, U, ρ

Step 5; Combine L, U, ρ with one additional variable, in sequence, to find the two pi products.



- Since F is the output, or dependent, variable, we select it to appear to the first power in the numerator:

$$\Pi_1 = L^a U^b \rho^c F = (L)^a (LT^{-1})^b (ML^{-3})^c (MLT^{-2}) = M^0 L^0 T^0$$

Equate exponents:

Length: $a + b - 3c + 1 = 0$

Mass: $c + 1 = 0$

Time: $-b - 2 = 0$

We can solve explicitly for

$$a = -2 \quad b = -2 \quad c = -1$$

Therefore $\Pi_1 = L^{-2} U^{-2} \rho^{-1} F = \frac{F}{\rho U^2 L^2} = C_f$ Ans.

- Finally, add viscosity to L , U , and ρ to find Π_1

$$\Pi_2 = L^a U^b \rho^c \mu^{-1} = L^a (LT^{-1})^b (ML^{-3})^c (ML^{-1}T^{-1})^{-1} = M^0 L^0 T^0$$

Equate exponents:

Length: $a + b - 3c + 1 = 0$

Mass: $c - 1 = 0$

Time: $-b + 1 = 0$

from which we find

$$a = b = c = 1$$

Therefore $\Pi_2 = L^1 U^1 \rho^1 \mu^{-1} = \frac{\rho UL}{\mu} = Re$



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Exercise: The power input P to a centrifugal pump is a function of the volume flow Q , impeller diameter D , rotational rate Ω , and the density ρ and viscosity μ of the fluid:

$$P = f(Q, D, \Omega, \rho, \mu)$$

Rewrite this as a dimensionless relationship. Hint: Use Ω , ρ , and D as repeating variables.

Exercise: At low velocities (laminar flow), the volume flow Q through a small-bore tube is a function only of the tube radius R , the fluid viscosity μ and the pressure drop per unit tube length dp/dx . Using the pi theorem, find an appropriate dimensionless relationship. $Q = f(\mu, R, dp/dx)$



- **Exercise;** Assume that the tip deflection δ of a cantilever beam is a function of the tip load P , beam length L , area moment of inertia I , and material modulus of elasticity E ; that is, $f(P, L, I, E)$. Rewrite this function in dimensionless form, and comment on its complexity and the peculiar value of j . (the scaling variable is 2)



An Alternate Step-by-Step Method by Ipsen

- The pi theorem method, just explained and illustrated, is often called the repeating variable method of dimensional analysis. Select the repeating variables, add one more, and you get a pi group. However, there are drawbacks:
 1. All pi groups contain the same repeating variables and might lack variety or effectiveness,
 2. one must (sometimes laboriously) check that the selected repeating variables do not form a pi group among themselves



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steps	Action Taken during This Step
1	Identify the significant dimensional variables and write out the primary dimensions of each.
2	Apply the Buckingham theorem to find the number of Pi-groups
3	Set up table with the number of rows equal to the number of dimensional variables and the number of columns equal to the number of basic dimensions plus one ($m + 1$) .
4	List all the dimensional variables in the first column with primary dimensions.
5	Select a dimension to be eliminated, choose a variable with that dimension in the first column, and combine with remaining variables to eliminate the dimension. List combined variables in the second column with remaining primary dimensions
6	Select another dimension to be eliminated, choose from variables in the second column that have that dimension, and combine with the remaining variables. List the new combinations with remaining primary dimensions in the third column.
7	Repeat Step 6 until all dimensions are eliminated.



Examples

1. There are three significant dimensional variables for a body falling in a vacuum (no viscous effects): the velocity V ; the acceleration due to gravity, g ; and the distance through which the body falls, h . Find the Pi-groups using the step-by-step method.

Solution;

A body is falling in a vacuum, $V = f(g, h)$

Step1; Significant variables and dimensions. $[V]=L/T$, $[g]=L/T^2$, $h=L$

Step 2; determine number of pi group. $n=3$, $j=2$, $k=3-2=1$ Pi group.

Step 3; Set up table with three rows (n) and three ($j+1$) columns.

Step 4; List variables and primary dimensions in first column.

Step 5; Select h to eliminate L . Divide g by h , enter in second column with dimension $1/T^2$. Divide V by h , enter in second column with dimension $1/T$.



- Example;** Let us illustrate with the same classical drag function proposed in Example 1 above. Underneath the variables, write out the dimensions of each quantity.

$$\begin{array}{ccccccc} F & = & \text{fcn} & (L, & V, & \rho, & \mu) \\ \{MLT^{-2}\} & & & \{L\} & \{LT^{-1}\} & \{ML^{-3}\} & \{ML^{-1}T^{-1}\} \end{array} \quad (5.1)$$

$$\begin{array}{ccccccc} \frac{F}{\rho} & = & \text{fcn} & \left(L, & V, & \cancel{\rho}, & \frac{\mu}{\rho} \right) \\ \{L^4T^{-2}\} & & & \{L\} & \{LT^{-1}\} & \{ML^{-3}\} & \{L^2T^{-1}\} \end{array}$$

$$\begin{array}{ccccccc} \frac{F}{\rho V^2} & = & \text{fcn} & \left(L, & \cancel{V}, & \frac{\mu}{\rho V} \right) \\ \{L^2\} & & & \{L\} & \{LT^{-1}\} & \{L\} \end{array}$$

$$\begin{array}{ccccccc} \frac{F}{\rho V^2 L^2} & = & \text{fcn} & \left(\cancel{L}, & \frac{\mu}{\rho VL} \right) \\ \{1\} & & & \{L\} & \{1\} \end{array}$$



- Recall the beam-deflection problem proposed in Example and the various dimensions:

$$\begin{array}{ccccccc} \delta & = & f(P, & L, & I, & E) \\ \{L\} & & \{MLT^{-2}\} & \{L\} & \{L^4\} & \{ML^{-1}T^{-2}\} \end{array}$$

$$\begin{array}{ccccccc} \delta & = & f\left(\frac{P}{E}, L, I, E\right) \\ \{L\} & & \{L^2\} & \{L\} & \{L^4\} & \{ML^{-1}T^{-2}\} \end{array}$$

$$\begin{array}{ccccccc} \frac{\delta}{L} & = & \text{fcn}\left(\frac{P}{EL^2}, \cancel{E}, \frac{I}{L^4}\right) \\ \{1\} & & \{1\} & \{L\} & \{1\} \end{array}$$

$$\frac{\delta}{L} = \text{fcn}\left(\frac{P}{EL^2}, \frac{I}{L^4}\right)$$



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- Step 6; Select g/h to eliminate T . Divide V/h by $\sqrt{g/h}$ and enter in third column.

Variables	[]	Variable	[]	Variable	[]
V	L/T	V/h	$1/T$	$(V/H)(h/g)^{1/2}$	---
g	L/T^2	g/h	$1/T^2$		
h	L	-	-		

$$\frac{V}{\sqrt{gh}} = \text{constant}$$



Modeling and Its Pitfalls

- Once the variables are selected and the dimensional analysis is performed, the experimenter seeks to achieve similarity between the model tested and the prototype to be designed.
- “Flow conditions for a model test are completely similar if all relevant dimensionless parameters have the same corresponding values for the model and the prototype.”

$$\Pi_1 = f(\Pi_2, \Pi_3, \dots, \Pi_k)$$

If $\Pi_{2m} = \Pi_{2p}$, $\Pi_{3m} = \Pi_{3p}$, Π_{1m} will equal Π_{1p} .



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Geometric Similarity; “A model and prototype are geometrically similar if and only if all body dimensions in all three coordinates have the same linear scale ratio.”

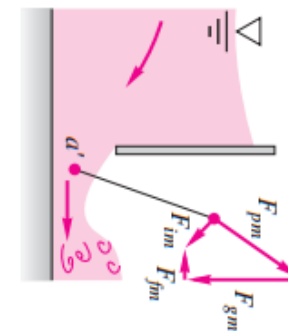
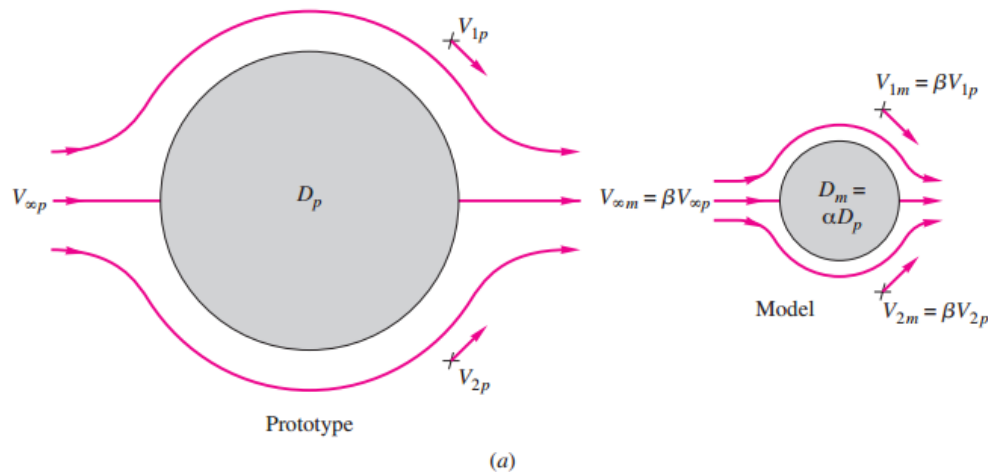
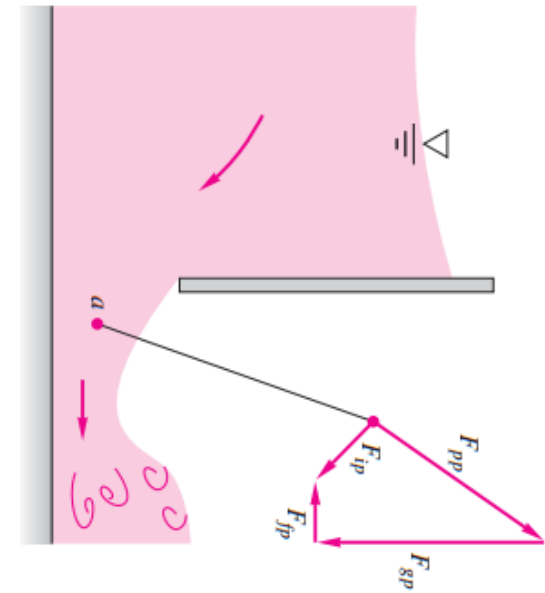
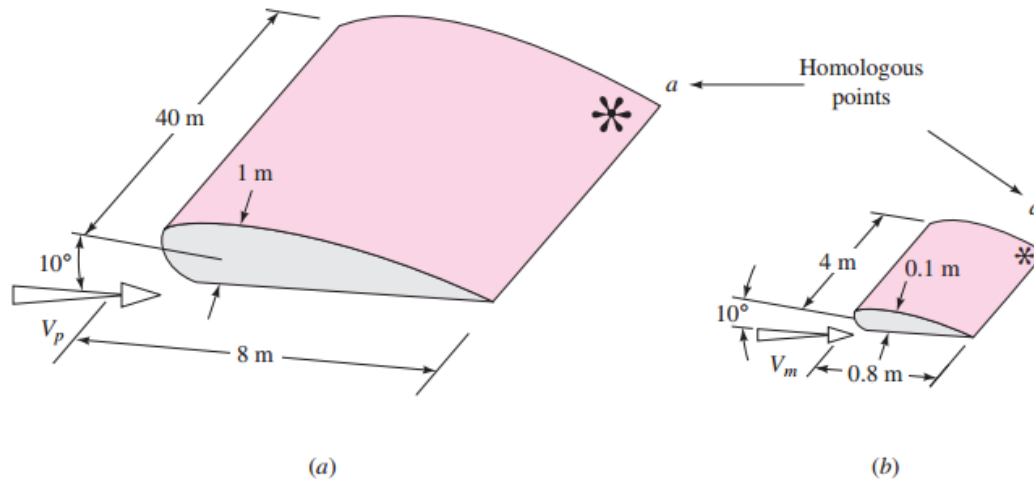
- All angles are preserved in geometric similarity. All flow directions are preserved. The orientations of model and prototype with respect to the surroundings must be identical.

Kinematic Similarity; Kinematic similarity requires that the model and prototype have the same length scale ratio and the same time scale ratio. The result is that the velocity scale ratio will be the same for both.

“The motions of two systems are kinematically similar if homologous particles lie at homologous points at homologous times.”



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Dynamic Similarity; Dynamic similarity exists when the model and the prototype have the same length scale ratio, time scale ratio, and force scale (or mass scale) ratio.

This is ensured if

1. For compressible flow, the model and prototype Reynolds number and Mach number and specific-heat ratio are correspondingly equal.
2. For incompressible flow
 - With no free surface: model and prototype Reynolds numbers are equal.
 - With a free surface: model and prototype Reynolds number, Froude number, and (if necessary) Weber number and cavitation number are correspondingly equal.



Dimensionless Groups in Fluid Mechanics

Parameter	Definition	Qualitative ratio of effects	Importance
Reynolds number	$Re = \frac{\rho UL}{\mu}$	$\frac{\text{Inertia}}{\text{Viscosity}}$	Almost always
Mach number	$Ma = \frac{U}{a}$	$\frac{\text{Flow speed}}{\text{Sound speed}}$	Compressible flow
Froude number	$Fr = \frac{U^2}{gL}$	$\frac{\text{Inertia}}{\text{Gravity}}$	Free-surface flow
Weber number	$We = \frac{\rho U^2 L}{Y}$	$\frac{\text{Inertia}}{\text{Surface tension}}$	Free-surface flow
Rossby number	$Ro = \frac{U}{\Omega_{\text{earth}} L}$	$\frac{\text{Flow velocity}}{\text{Coriolis effect}}$	Geophysical flows
Cavitation number (Euler number)	$Ca = \frac{p - p_v}{\rho U^2}$	$\frac{\text{Pressure}}{\text{Inertia}}$	Cavitation
Prandtl number	$Pr = \frac{\mu c_p}{k}$	$\frac{\text{Dissipation}}{\text{Conduction}}$	Heat convection
Eckert number	$Ec = \frac{U^2}{c_p T_0}$	$\frac{\text{Kinetic energy}}{\text{Enthalpy}}$	Dissipation
Specific-heat ratio	$k = \frac{c_p}{c_v}$	$\frac{\text{Enthalpy}}{\text{Internal energy}}$	Compressible flow



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Strouhal number	$St = \frac{\omega L}{U}$	$\frac{\text{Oscillation}}{\text{Mean speed}}$	Oscillating flow
Roughness ratio	$\frac{\epsilon}{L}$	$\frac{\text{Wall roughness}}{\text{Body length}}$	Turbulent, rough walls
Grashof number	$Gr = \frac{\beta \Delta T g L^3 \rho^2}{\mu^2}$	$\frac{\text{Buoyancy}}{\text{Viscosity}}$	Natural convection
Rayleigh number	$Ra = \frac{\beta \Delta T g L^3 \rho^2 c_p}{\mu k}$	$\frac{\text{Buoyancy}}{\text{Viscosity}}$	Natural convection
Temperature ratio	$\frac{T_w}{T_0}$	$\frac{\text{Wall temperature}}{\text{Stream temperature}}$	Heat transfer
Pressure coefficient	$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U^2}$	$\frac{\text{Static pressure}}{\text{Dynamic pressure}}$	Aerodynamics, hydrodynamics
Lift coefficient	$C_L = \frac{L}{\frac{1}{2} \rho U^2 A}$	$\frac{\text{Lift force}}{\text{Dynamic force}}$	Aerodynamics, hydrodynamics
Drag coefficient	$C_D = \frac{D}{\frac{1}{2} \rho U^2 A}$	$\frac{\text{Drag force}}{\text{Dynamic force}}$	Aerodynamics, hydrodynamics
Friction factor	$f = \frac{h_f}{(V^2/2g)(L/d)}$	$\frac{\text{Friction head loss}}{\text{Velocity head}}$	Pipe flow
Skin friction coefficient	$c_f = \frac{\tau_{\text{wall}}}{\rho V^2/2}$	$\frac{\text{Wall shear stress}}{\text{Dynamic pressure}}$	Boundary layer flow



**Thank you
end of chapter five**